```
clc;
clear;
close all;

% WIFI SIGNAL STRENGTH AND NETWORK ANALYSIS

% Configuration
targetSSID = 'YOUR_WIFI_SSID';
numSamples = 10;
channels = 1:14;

% Sample measurements
% Replace these values with actual measured data
timeSeconds = (1:numSamples)' * 2;

targetRSSI = [-55 -57 -54 -60 -58 ...
              -56 -59 -53 -55 -57]';

targetChannel = [6 6 6 6 6 6 6 6 6 6]';

visibleAPs = [8 9 8 10 9 8 11 9 8 10]';

averagePing = [4.5 5.2 4.8 6.1 5.0 ...
               4.7 5.5 4.4 4.9 5.1]';

packetLoss = [0 0 0 5 0 0 0 0 0 0]';

% Channel counts for each sample
% Rows represent samples; columns represent channels 1 to 14
channelCounts = zeros(numSamples, 14);

% Example channel distribution
channelCounts(:,1) = [2 2 2 3 2 2 3 2 2 3]';
channelCounts(:,6) = [3 3 2 3 3 2 3 3 2 3]';
channelCounts(:,11) = [3 4 4 4 4 4 5 4 4 4]';

% Display measurement table
sampleNumber = (1:numSamples)';

resultsTable = table(sampleNumber, timeSeconds, ...
    targetRSSI, targetChannel, visibleAPs, ...
    averagePing, packetLoss, ...
    'VariableNames', ...
    {'Sample','Time_s','RSSI_dBm','TargetChannel', ...
     'VisibleAPs','AveragePing_ms','PacketLoss_percent'});

disp('WIFI MEASUREMENT RESULTS');
disp(resultsTable);

% Plot 1: Wi-Fi signal strength
figure;
plot(timeSeconds, targetRSSI, 'bo-', 'LineWidth', 1.5);
```

```
grid on;
xlabel('Time (seconds)');
ylabel('Signal Strength (dBm)');
title('Wi-Fi RSSI versus Time');

% Plot 2: Visible access points
figure;
plot(timeSeconds, visibleAPs, 'rs-', 'LineWidth', 1.5);
grid on;
xlabel('Time (seconds)');
ylabel('Number of Visible Access Points');
title('Visible Wi-Fi Access Points versus Time');

% Plot 3: Channel distribution
figure;
bar(channels, sum(channelCounts, 1));
grid on;
xlabel('Wi-Fi Channel');
ylabel('Total Access Point Observations');
title('Wi-Fi Channel Distribution');
xlim([0 15]);
xticks(1:14);

% Plot 4: Average ping latency
figure;
plot(timeSeconds, averagePing, 'md-', 'LineWidth', 1.5);
grid on;
xlabel('Time (seconds)');
ylabel('Average Ping Latency (ms)');
title('Gateway Ping Latency versus Time');

% Plot 5: Packet loss
figure;
stem(timeSeconds, packetLoss, 'filled');
grid on;
xlabel('Time (seconds)');
ylabel('Packet Loss (%)');
title('Gateway Packet Loss versus Time');

% Calculate average measurements
meanRSSI = mean(targetRSSI);
meanVisibleAPs = mean(visibleAPs);
meanPing = mean(averagePing);
meanPacketLoss = mean(packetLoss);

fprintf('\nWIFI ANALYSIS SUMMARY\n');
fprintf('------------------------------\n');
fprintf('Target Wi-Fi SSID: %s\n', targetSSID);
fprintf('Average RSSI: %.2f dBm\n', meanRSSI);
fprintf('Average visible APs: %.2f\n', meanVisibleAPs);
fprintf('Average gateway ping: %.2f ms\n', meanPing);
fprintf('Average packet loss: %.2f percent\n', meanPacketLoss);

fprintf('\nAnalysis completed successfully.\n');
```

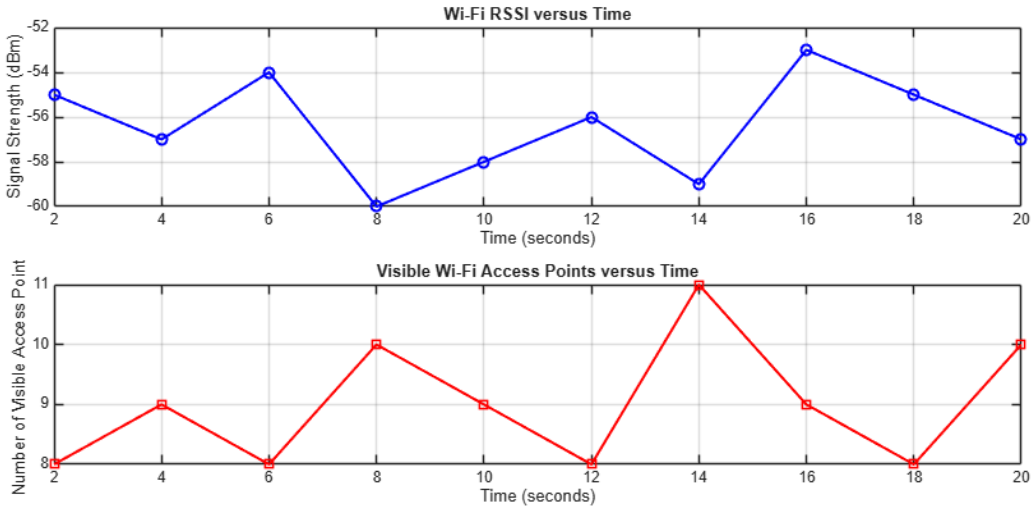
WIFI MEASUREMENT RESULTS

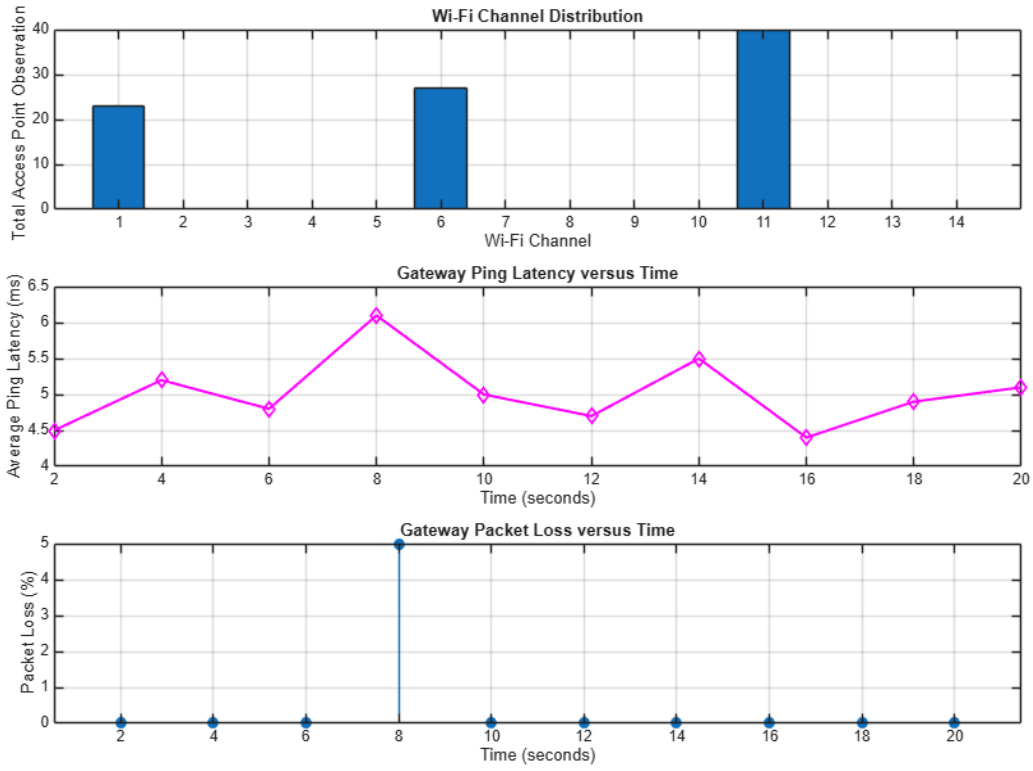
Sample	Time_s	RSSI_dBm	TargetChannel	VisibleAPs	AveragePing_ms	PacketLoss_percent
1	2	-55	6	8	4.5	0
2	4	-57	6	9	5.2	0
3	6	-54	6	8	4.8	0
4	8	-60	6	10	6.1	5
5	10	-58	6	9	5	0
6	12	-56	6	8	4.7	0
7	14	-59	6	11	5.5	0
8	16	-53	6	9	4.4	0
9	18	-55	6	8	4.9	0
10	20	-57	6	10	5.1	0

WIFI ANALYSIS SUMMARY

```
------------------------------
Target Wi-Fi SSID: YOUR_WIFI_SSID
Average RSSI: -56.40 dBm
Average visible APs: 9.00
Average gateway ping: 5.02 ms
Average packet loss: 0.50 percent

Analysis completed successfully.
```





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